5. Straight line joining the points $(2,3)$ and $(-1,4)$ passes through the point (α,β) if

   (1) $\alpha+2\beta=7$ (2) $3\alpha+\beta=9$ (3) $\alpha+3\beta=11$ (4) $3\alpha+\beta=11$

6. The slope of the line which makes an angle 45° with the line $3x-y=-5$ are

   (1) 1, -1 (2) $\frac{1}{2}$, -2 (3) 1, $\frac{1}{2}$ (4) 2, $-\frac{1}{2}$

7. Equation of the straight line that forms an isosceles triangle with coordinate axes in the I-quadrant with perimeter $4+2\sqrt{2}$ is

   (1) $x+y+2=0$ (2) $x+y-2=0$ (3) $x+y-\sqrt{2}=0$ (4) $x+y+\sqrt{2}=0$

8. The coordinates of the four vertices of a quadrilateral are (−2,4), (−1,2), (1,2) and (2,4) taken in order.The equation of the line passing through the vertex (−1,2) and dividing the quadrilateral in the equal areas is

   (1) $x+1=0$ (2) $x+y=1$ (3) $x+y+3=0$ (4) $x-y+3=0$

9. The intercepts of the perpendicular bisector of the line segment joining (1, 2) and (3,4) with coordinate axes are

   (1) 5, -5 (2) 5, 5 (3) 5, 3 (4) 5, -4

10. The equation of the line with slope 2 and the length of the perpendicular from the origin equal to $\sqrt{5}$ is

    (1) $x-2y=\sqrt{5}$ (2) $2x-y=\sqrt{5}$ (3) $2x-y=5$ (4) $x-2y-5=0$

11. A line perpendicular to the line $5x-y=0$ forms a triangle with the coordinate axes. If the area of the triangle is 5 sq. units, then its equation is

    (1) $x+5y\pm5\sqrt{2}=0$ (2) $x-5y\pm5\sqrt{2}=0$ (3) $5x+y\pm5\sqrt{2}=0$ (4) $5x-y\pm5\sqrt{2}=0$

12. Equation of the straight line perpendicular to the line$x-y+5=0$, through the point of intersection the y-axis and the given line

    (1) $x-y-5=0$ (2) $x+y-5=0$ (3) $x+y+5=0$ (4) $x+y+10=0$

13. If the equation of the base opposite to the vertex (2, 3) of an equilateral triangle is $x+y=2$, then the length of a side is

    (1) $\sqrt{\frac{3}{2}}$ (2) 6 (3) $\sqrt{6}$ (4) $3\sqrt{2}$

14. The line $(p+2q)x+(p-3q)y=p-q$ for different values of p and q passes through the point

    (1) $\left(\frac{3}{2},\frac{5}{2}\right)$ (2) $\left(\frac{2}{5},\frac{2}{5}\right)$ (3) $\left(\frac{3}{5},\frac{3}{5}\right)$ (4) $\left(\frac{2}{5},\frac{3}{5}\right)$

15. The point on the line $2x-3y=5$ is equidistance from $(1,2)$ and $(3,4)$ is

    (1) $(7,3)$ (2) $(4,1)$ (3) $(1,-1)$ (4) $(-2,3)$

16. The image of the point (2, 3) in the line $y=-x$ is

    (1) $(-3,-2)$ (2) $(-3,2)$ (3) $(-2,-3)$ (4) $(3,2)$

17. The length of $\perp$ from the origin to the line $\frac{x}{3}-\frac{y}{4}=1$, is

    (1) $\frac{11}{5}$ (2) $\frac{5}{12}$ (3) $\frac{12}{5}$ (4) $-\frac{5}{12}$

18. The y-intercept of the straight line passing through (1,3) and perpendicular to $2x-3y+1=0$ is

    (1) $\frac{3}{2}$ (2) $\frac{9}{2}$ (3) $\frac{2}{3}$ (4) $\frac{2}{9}$